Equilibrium or Retrieval? Do LLM Agents Derive or Recall Game-Theoretic Equilibria, and What That Means for Mechanism Design

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Abstract

Large language models (LLMs) play textbook games well, but a model can reach the textbook answer in two ways: by deriving it from the rules, or by recognising a famous game and recalling its answer. We separate the two with *isomorphic twins*: every question is asked in its standard wording, in a disguised wording with identical numbers and rules but no game vocabulary, and, in a third pre-registered study, in an extreme disguise with no story, coded answers and a neutral answer box. Four models from four providers answer 18,768 questions across a Vickrey auction, a Cournot duopoly, a finitely repeated prisoner's dilemma (PD) and a public goods game, each checked by a competence probe. All three studies are pre-registered on OSF. We fit a logit quantal response equilibrium (QRE) to every cell and compare precision λ across wordings. For the first model analysed (Nemotron 3 Ultra), the standard disguise changed nothing: play was identical in both wordings, and the model named the disguised game in 54–100% of answers, so the disguise did not hide the game. [TBD: 2A results for the other three models; Stage 2C (extreme disguise) results.] We then ask what noisy, QRE-like agents imply for mechanism design. We prove that in a second-price auction a payment multiplier α is irrelevant for QRE bidders, but not for bidders who recall "bid your value", so α separates derivation from retrieval. We show that the revenue-optimal reserve for QRE bidders is U-shaped in λ, but gains at most about 1% over Myerson's reserve. And we derive in closed form the cooperation subsidy that reaches any target cooperation rate in a PD. In the stage game, its halfway point is free of parameters for the whole regular QRE family. A pre-registered test compares the derived subsidy's predictions with a classically "guessed" one: [TBD: Stage 5 result].

1 Introduction

LLM agents are beginning to take part in settings that game theory was built to analyse: bidding, pricing, negotiating, and sharing common resources. Designing the rules for such settings, which is the job of mechanism design, needs a model of how the participants behave. Classical mechanism design assumes rational players who best-respond. Behavioural mechanism design replaces them with an empirically fitted model, such as the logit quantal response equilibrium [6, 11]. Before a fitted model of LLM play can be trusted to design anything, a prior question must be answered: *is the play we measure reasoning at all, or retrieval?*

The concern is concrete. The Vickrey auction, the Cournot duopoly, the prisoner's dilemma and the public goods game are among the most written-about games in any training corpus. A model can answer "bid your true value" or "produce $(a-c)/3b$" because it has seen the answer thousands of times, not because it derived it. LLMs are known to perform better on familiar versions of a task than on unfamiliar but equivalent ones [10]. If strategic play is recalled, then a mechanism

tuned to it may fail as soon as the rules depart from the textbook, which is exactly when a designer would change them.

What we do.

1. **Isomorphic twins.** Each question has a standard ("named") wording and a disguised wording. The disguise keeps every number and rule but removes the game's vocabulary. A third study adds an *extreme* disguise: no story, coded answers (for example, offer $= 3x + 5$) and a neutral answer box. A model that derives should play identically across wordings; a model that recalls should not.

2. **Competence probes and pre-registration.** Every answer is paired with a short maths check posed in the question's own terms, and only answers whose check is passed are analysed. All hypotheses, sample sizes and analyses were pre-registered on OSF before data collection, in three registrations [13].

3. **Behavioural fits.** We fit a logit QRE to each model, game and wording, and compare precision λ across wordings ($\Delta\lambda$).

4. **Mechanism design for noisy agents.** We derive three results for QRE players: that a payment multiplier is irrelevant in a second-price auction; that a reserve price can only lower efficiency, while the revenue-optimal reserve is U-shaped in λ; and a closed-form cooperation subsidy for the PD. We then test the derived subsidy on the models against a classically guessed one.

Contributions. (i) A pre-registered, probe-checked twin design that separates derived from recalled strategic play. (ii) Evidence on four frontier models across four games and three wordings [TBD: one-sentence headline once all models are analysed]. (iii) Theory results for mechanism design with QRE agents, including a parameter-free prediction (Corollary 1) and an α-based test of retrieval (Theorem 1), with a confirmatory test on the same models.

2 Related work

LLMs as economic and strategic agents. Horton et al. [8] proposed treating LLMs as simulated economic agents, "homo silicus", and reproduced classic behavioural-economics experiments with them. Akata et al. [2] studied LLMs in finitely repeated 2×2 games, including the PD, and found strong play in self-interested games but poor coordination. Fan et al. [5] evaluated whether LLMs satisfy basic rationality requirements in games, and benchmarks such as GTBench [4] measure strategic reasoning across many game types. Our design differs: we hold the game fixed and vary only its *wording*, so that a difference in play isolates recognition of the game rather than ability.

Behavioural game theory for LLMs. Jia et al. [9] evaluated 22 LLMs with behavioural game-theory models rather than Nash benchmarks. Pechon-Elkins and Chun [14] audit QRE-based measures of LLM strategic reasoning and argue that fitted parameters are payoff- and model-dependent summaries rather than structural measures of rationality. We take that caution seriously: we report fits per game rather than pooling one λ; we verify by parameter recovery that our fitting procedure recovers known λ; and we use λ mainly for *within-question* comparisons across wordings, where the payoffs are identical by construction.

Familiarity versus reasoning. McCoy et al. [10] showed that LLM accuracy depends on how probable a task and its output are under the training distribution. Our twins apply this idea to strategic play, with the important addition of competence probes, so that a drop under disguise cannot simply reflect misreading the disguised rules.

Mechanism design. Our auction results build on the second-price auction [16], on Myerson's optimal auction and implementability results [12], and on the treatment in Roughgarden [15]. QRE was introduced by McKelvey and Palfrey [11] and is surveyed in Goeree et al. [6]. To our knowledge, the closed forms and invariance results in Section 6 are not stated in this form for LLM agents elsewhere.

3 Games, twins and competence probes

Games. Four games with known, unique equilibrium actions:

- **Vickrey auction:** three bidders; private value uniform on $[10, 100]$; the equilibrium is to bid one's value.
- **Cournot duopoly:** inverse demand $P = a - bQ$ with $b = 1$ and $c = 10$, and a drawn per question; equilibrium quantity $(a - c)/3b$.
- **Repeated PD:** 10 rounds, a horizon known to both players, payoffs $(T, R, P, S) = (5, 3, 1, 0)$, asked in round 1; backward induction gives defection throughout.
- **Public goods:** n players, endowment e, multiplier $1 < M < n$; the equilibrium is to contribute nothing.

Numbers are drawn from seeded samplers, so each question can be reproduced exactly.

Twins. The disguised wording keeps every number, rule and information structure (who knows what, and who moves when) and removes the game's vocabulary ("auction", "firm", "cooperate"). The PD's labels are counterbalanced, with `X`/`Y` randomly assigned to cooperation. Each wording has three paraphrases, which reword or reorder but never add or drop a fact, so a wording effect can be told apart from a disguise effect. The *extreme* disguise (Stage 2C) goes further:

- no story at all;
- coded answers (Vickrey offer $= 3x + 5$ with a private prize; Cournot amount $= 2x + 4$ with the profit written in expanded form; public goods asks how much to *keep*);
- PD payoffs transformed to $2u - 4$ and shown as balance changes;
- a neutral answer box.

Each extreme question uses the same seed and paraphrase index as its standard twin, so every question exists in three wordings with identical numbers.

Competence probes. After each answer, the model receives a short calculation in the same wording (for example, "you bid 50, the highest other bid is 60; what is your payoff?"). Only answers whose probe is passed are analysed. Verdicts are AT_EQUILIBRIUM, STRATEGIC_GAP (probe passed, equilibrium missed), COMPETENCE_GAP (probe failed), and parse or format failures. Server refusals are infrastructure failures and are re-asked, never scored.

Models. gpt-oss-120b (via Groq), Nemotron 3 Ultra (NVIDIA), Kimi K3 (Moonshot, via NVIDIA) and Gemini 3.5 Flash Lite (Google), all with default sampling, collected September–November 2026.

Studies. Three pre-registered studies, 18,768 questions in total:

- **Stage 2A** (pre-registration 1): 4 games × 2 wordings × 138 questions × 4 models = 4,416.
- **Stage 2B** (pre-registration 2): 12,144 questions in three blocks:
  - *traps*, variants whose textbook answer is wrong (for example, the winner pays *twice* the second bid, or there are three firms instead of two);
  - a *number sweep* that varies one parameter;
  - a PD *endgame* asked at rounds 5, 9 and 10 after mutual cooperation.
- **Stage 2C** (pre-registration 3): the extreme disguise, 2,208 questions.

4 Pre-registration and analysis

Pre-registration. Pre-registrations 1–3 were locked by SHA-256 fingerprint in git and registered on OSF before the corresponding data existed. Each fixes:

- the predictions, written by the author;
- the exact sample sizes, as a stopping rule;
- exclusions;
- the tests.

The analysis code was fingerprinted too, and it refuses to run on real data unless its fingerprint matches. Ten deviations, all found and recorded before the affected analysis, are listed in Appendix A.

Main tests.

- **Stage 2A:** equilibrium play, named vs disguised, two-proportion z-tests with Holm correction across games [7]. Public goods, the predicted null, uses an equivalence test (TOST, ±13 points).
- **Stage 2C:** each extreme question is paired with its normal and disguised twins; exact McNemar tests, paired differences with Agresti–Min intervals [1], and pooling across models by summing discordant pairs (the Mantel–Haenszel test for matched pairs).

QRE fitting. In a logit QRE a player chooses action a with probability proportional to $\exp(\lambda\, u(a))$, where u is the expected payoff against the others' equilibrium behaviour. $\lambda = 0$ is uniform noise and $\lambda \to \infty$ is Nash play. The game-specific choices were fixed before fitting:

- the PD and Cournot use symmetric fixed points;
- in public goods, $u(g) = (M/n - 1)g$;

- in the Vickrey auction, rivals are assumed to bid their equilibrium bids, with values uniform on $[10, 100]$.

We compute a posterior on a log-uniform grid $\lambda \in [10^{-3}, 10^{3}]$. A cell whose evidence keeps rising towards the top of the grid gets a lower bound only, and is flagged as a ceiling. Parameter recovery on simulated data gave 93.75% coverage of nominal 95% intervals across 320 fits.

Reasoning codes. The models' stated reasoning is coded by a fixed, locked codebook. The key code is NAMES_GAME: the reasoning names the game, using a name that the question itself does not use. Agreement with blind hand coding by the author was $\kappa = 1.00$ on Stage 2A wording, and again on all 40 answers of a blind sample in the extreme wording.

The fraction slip. With a number-only answer box, Nemotron sometimes derives a fraction ($683/60 \approx 11.38$) and writes only part of it (683) into the box. A mechanical correction rule was fixed in pre-registration 3, before any Stage 2C data, and is applied identically to both stages. It changed 25 of Nemotron's 276 Stage 2A Cournot answers, all to correct, and no answer from any other model. Pre-registration 1's own analysis is reported uncorrected, as registered.

5 Results

5.1 Stage 2A: standard vs disguised wording

Table 1: Equilibrium play, named vs disguised (Stage 2A, pre-registration 1), share of probe-passed answers.

Model	Game	Named	Disguised	Change	Holm p
Nemotron	Vickrey	100% (138/138)	100% (138/138)	+0	1
	Cournot	90% (124/138)	88% (121/138)	−2	1
	Repeated PD	59% (81/137)	59% (81/137)	+0	1
	Public goods	100% (136/136)	100% (138/138)	+0	1
gpt-oss-120b	all games	[TBD: collection ends about late October]			
Kimi K3	all games	[TBD:]			
Gemini 3.5 Flash Lite	all games	[TBD:]			

For Nemotron, the disguise changed nothing (Table 1). Public goods was confirmed equivalent within ±13 points (TOST $p < 10^{-30}$). The pre-registered prediction of a Cournot drop was not borne out for this model. The fitted precision tells the same story:

- Vickrey and public goods are at the ceiling in both wordings ($\lambda \geq 944$ and $\lambda \geq 21.8$), so no difference is detectable.
- The PD gives $\lambda = 0.154$ $[0.0016, 0.47]$ named against 0.188 $[0.0018, 0.50]$ disguised, $P(\Delta\lambda > 0) = 0.44$.

The wording-free reading of these results, however, is not "Nemotron derives". The reasoning codes show why.

5.2 The disguise did not hide the game

Nemotron *named* the disguised game in 100% (Vickrey), 96% (Cournot), 81% (PD) and 54% (public goods) of its disguised answers. A disguise that the model sees through cannot test derivation against recall. Two causes were found:

- the disguised answer box still used the named game's field names ("bid", "quantity", "contribution"), recorded as deviation 8;
- strong models recognise the mathematical structure itself.

This motivated the extreme disguise. In a pilot on practice questions, it cut Nemotron's naming of Cournot from 96% to 14% and of the PD from 81% to 50%, while maths-check pass rates were unchanged.

Why the equilibrium was missed. The misses are informative:

- **Cournot:** every named-wording miss, and 14 of 17 disguised misses, was a fraction slip, i.e. correct reasoning written into the box incorrectly.
- **PD:** every named-wording cooperative choice gave a cooperation-norm reason (trust, mutual benefit), even though 91% of PD answers mentioned backward induction.

So Nemotron's departures from equilibrium are not symmetric noise. They are formatting errors (Cournot) and a systematic norm (PD).

5.3 Stage 2B: traps, sweeps and the endgame

[TBD: Block B (traps): correct share and the "recall signature" (textbook answer given where it is wrong), named vs disguised, CMH across models. Block C (sweep): wording × size interaction. Block E (endgame): defection at rounds 5, 9, 10. Analysis is locked; it runs when all four models are complete.]

5.4 Stage 2C: the extreme disguise

[TBD: Per model and game: equilibrium play, normal vs extreme (McNemar, Holm), the pooled test, recognition extreme vs disguised, conversion slips, decoded λ, and the mechanical check of each pre-registered prediction (for example, Cournot drops of at least 35 points for Nemotron, 20–35 for Gemini, and at most 15 for Kimi and gpt-oss). The analysis is locked; it runs as each model completes.]

6 Mechanism design for noisy agents

We now treat each player as a logit QRE player with precision λ and ask how the rules should change. Proofs of Theorems 1–3 were worked through by the author (see the contributions statement).

6.1 Second-price auctions with a reserve and a payment multiplier

Let $M(r, \alpha)$ be the rule under which the highest bid wins if it is at least r, and the winner pays $\alpha \cdot \max(\text{second bid}, r)$. $M(0, 1)$ is the Vickrey auction. Under $M(0, \alpha)$ the dominant strategy is $b = v/\alpha$.

Theorem 1 (α is irrelevant for QRE bidders)**.** *If $(B_1, \ldots, B_n)$ is a QRE of $M(r, 1)$ on bid space $[0, \bar{B}]$, then $(B_1/\alpha, \ldots, B_n/\alpha)$ is a QRE of $M(r/\alpha, \alpha)$ on $[0, \bar{B}/\alpha]$, with the same winner, revenue and payoffs.*

Proof. Against a highest rival bid H, $U_\alpha(b \mid v) = \mathbb{E}[(v - \alpha H)\mathbf{1}\{H < b\}]$. With $H = H_1/\alpha$ and $b = \beta/\alpha$, $U_\alpha(\beta/\alpha \mid v) = U_1(\beta \mid v)$. So the M_α bid has density $\propto \exp(\lambda U_1(\alpha b \mid v))$, and by a change of variables, αB has the M_1 QRE density. The scaled profile is therefore self-consistent. Dividing every bid by α keeps their order, and revenue is $\alpha \cdot B_{(2)}/\alpha = B_{(2)}$. □

Consequence: α is a test of retrieval. For a deriving QRE bidder, $\alpha \cdot b$ has the same distribution at every α. A bidder that recalls "bid your value" bids v whatever α is. With three such bidders under $\alpha = 2$, the winner expects $\mathbb{E}[v_{(1)}] - 2\,\mathbb{E}[v_{(2)}] = 77.5 - 110 = -32.5$. Stage 2B's "pays twice" trap is exactly this test: [TBD: trap result].

Theorem 2 (a reserve never raises efficiency)**.** *For any λ, expected efficiency (the winner's value, or 0 if there is no sale) cannot be increased by $r > 0$.*

Proof. For fixed bids, r never changes who has the highest bid; it can only cancel the sale, which replaces a value of at least 10 by 0. Under QRE the bids also respond to r. Solving the full symmetric QRE numerically, efficiency falls in r at every λ tested. □

So the originally planned problem, maximising efficiency over (r, α), has a trivial answer. For *revenue*, rational bidders with values uniform on $[10, 100]$ have virtual value $\varphi(v) = 2v - 100$ and Myerson's reserve $r^* = 50$. Table 2 gives $r^*(\lambda)$ for QRE bidders, from the full symmetric QRE (each bidder responding to the others' noisy bids).

Table 2: Revenue-optimal reserve for three logit-QRE bidders (values uniform on $[10, 100]$).

λ	$r^*(\lambda)$	Revenue at r^*	Revenue at 50
0.02	57.0	69.89	69.64
0.1	44.0	58.68	58.37
0.2	41.5	56.15	55.60
1	45.0	55.11	54.98
5	48.5	56.09	56.08
100	49.5	56.67	56.67

$r^*(\lambda)$ is U-shaped (Figure 1):

- **Moderate noise:** 21% of bidders with $v > 50$ bid below a reserve of 50 (at $\lambda = 0.2$), so lowering r recovers those sales.
- **Extreme noise:** bids barely track values (a bidder worth 20 bids 59 on average at $\lambda = 0.02$), and the reserve becomes a floor under random overbids.

The gain over Myerson's reserve is at most about 1% of revenue. And for Nemotron, whose auction play is at the ceiling, there is nothing to correct. The auction is therefore not where a behaviourally derived mechanism can matter for these models.

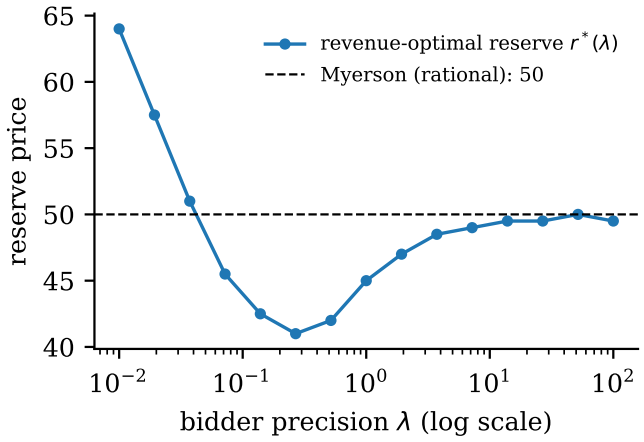


Figure 1: The revenue-optimal reserve for three logit-QRE bidders (full symmetric QRE, values uniform on $[10, 100]$). It is U-shaped in precision, and returns to Myerson's 50 as $\lambda \to \infty$.

6.2 A cooperation subsidy for the prisoner's dilemma

Under rule $S(\tau)$, every player who cooperates is paid an extra τ each round. Against an opponent who cooperates with probability p, defecting pays more by $\Delta U(p) = p(T - R) + (1 - p)(P - S)$. Under $S(\tau)$ this advantage becomes $\Delta U(p) - \tau$, and the symmetric QRE satisfies $p = 1/(1 + \exp(\lambda(\Delta U(p) - \tau)))$.

Theorem 3 (closed-form subsidy)**.** *The subsidy at which the symmetric logit QRE cooperation rate equals p^* is*

$$\tau(p^*, \lambda) = \Delta U(p^*) + \frac{\operatorname{logit}(p^*)}{\lambda}.$$

Since $d\tau/dp = (T - R) - (P - S) + 1/(\lambda p(1 - p)) > 0$, each τ gives a unique QRE.

Proof. Rearrange: $(1 - p)/p = \exp(\lambda(\Delta U(p) - \tau))$, so $-\operatorname{logit}(p) = \lambda(\Delta U(p) - \tau)$. □

Corollary 1 (a parameter-free halfway point)**.** *At $\tau_{1/2} = \Delta U(\frac{1}{2}) = \frac{1}{2}[(T - R) + (P - S)]$ (that is, 1.5 for our payoffs), every regular QRE population cooperates with probability exactly $\frac{1}{2}$, whatever its precision and whatever the (symmetric) error distribution.*

Proof. An indifferent regular quantal responder randomises evenly. Uniqueness follows because $\Delta U(p) - \tau$ rises in p while the response falls in it. □

Rational players defect for $\tau < P - S = 1$ and cooperate for $\tau > T - R = 2$. In between, the subsidised stage game is chicken, whose symmetric equilibrium mixes with $p = \tau - 1$, so it too passes through $\frac{1}{2}$ at $\tau = 1.5$. The corollary is a statement about the *stage* game. Chicken has two pure equilibria that pay each player differently, so in the 10-round game round-1 cooperation can be subgame-perfect: play (C, C) and punish a deviator in the last round [3]. At $\tau = 1.5$ deviating gains 0.5, while the last-round punishment costs at least 1.5. Cooperation above 50% at $\tau = 1.5$ is therefore consistent with rational repeated-game play, as well as with a cooperation norm. Noisy players are cheap to nudge towards partial cooperation and expensive to push to near-certainty (Figure 2).

At Nemotron's fitted $\lambda = 0.154$:

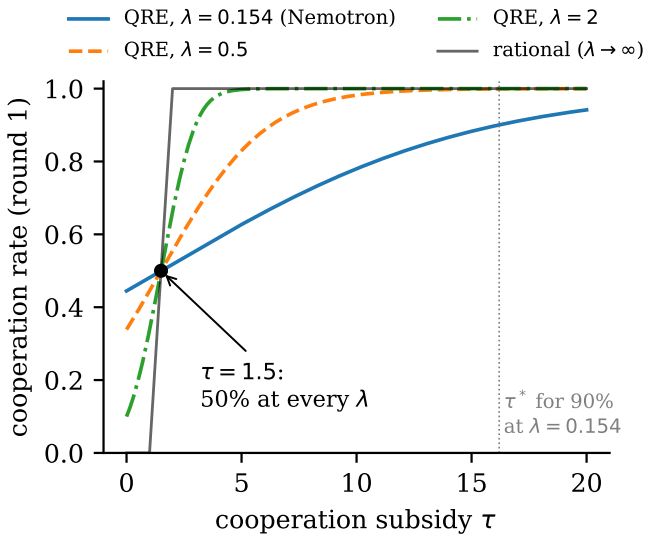


Figure 2: Cooperation under a subsidy τ for logit-QRE players. Every precision passes through 50% at $\tau = 1.5$ (Corollary 1), as does the rational mixed equilibrium. The dotted line is Theorem 3's subsidy for 90% at Nemotron's fitted $\lambda = 0.154$.

- $\tau = 1.5$ gives 50%;
- $\tau = 2.5$, the "classical" choice just above 2, gives only about 54%;
- 90% needs $\tau \approx 16.2$.

7 A confirmatory test of the derived mechanism

The original plan's headline comparison is a mechanism derived from the fitted behaviour against a naively guessed one. Stage 5 asks each model the Stage 2A PD (round 1, standard wording) with the subsidy stated as a separate rule for both players, at four levels:

- $\tau = 0$ (baseline);
- $\tau = 1.5$ (Corollary 1);
- $\tau = 2.5$ (the classical guess);
- $\tau^*(0.9, \hat{\lambda}_m)$, Theorem 3's subsidy for 90% cooperation at model m's own fitted λ (the derived mechanism).

There are 100 questions per level per model, balanced across the three paraphrases, each with a maths check that requires applying the subsidy. The design, predictions and tests will be pre-registered (pre-registration 4) before any subsidy question is asked.

"More subsidy gives more cooperation" would test nothing: $\tau^* = 1.9 + \ln 9/\lambda > 2.5$ whenever $\lambda < 3.66$. The derived-vs-guessed comparison is therefore between the two theories' *predictions*:

- at $\tau = 2.5$, is QRE's prediction closer to the observed rate than classical theory's 100%?

- at τ^*, is the observed rate within $90 \pm 15\%$?

The stage-game halfway point at $\tau = 1.5$ is read with the repeated-game caveat above. The reasoning codes separate a cooperation norm (fairness) from repeated-game reasoning (future rounds, reciprocity).

[TBD: Results: cooperation by level and model; derived vs guessed; the halfway-point test.]

8 Limitations

- **Disguises can fail.** The standard disguise did not hide the games from the first model analysed. The extreme disguise hides some games (Cournot) better than others (Vickrey and public goods, whose rules are their fingerprint), so a null result under a disguise the model sees through says nothing about derivation.
- **Timing.** Stage 2C was collected weeks after Stage 2A, so a provider updating a model in between would show up as a difference.
- **The behaviour model.** QRE treats every departure from equilibrium as payoff-scaled noise. Nemotron's departures are not: they are formatting slips and a cooperation norm. Fitted λ should therefore be read as a summary, not as a structural parameter [14].
- **The stage-game QRE in the PD.** It ignores continuation play in the 10-round game. That is harmless for rational players when the stage equilibrium is unique ($\tau = 0$, 2.5, τ^*). It is not harmless at $\tau = 1.5$, where round-1 cooperation can be subgame-perfect.
- **Four models.** With four models, cross-model patterns are rankings, not correlations.
- **Free tiers.** All models were accessed through free tiers with default sampling. Provider-side changes, rate limits and outages were logged, and outages were re-asked rather than scored (Appendix A).

9 Conclusion

[TBD: Two paragraphs, once results are in: what the twins show about derive vs recall for each model, and whether the theory-derived subsidy beat the guessed one.]

Author contributions and use of AI

This project was conducted by the author, a sixth-form student, with extensive assistance from Claude (Anthropic), an AI model.

The author:

- amended the Stage 5 design (the headline test, and the repeated-game caveat at $\tau = 1.5$), fixed the decision-gate criteria, and fixed the interpretation of each possible result in advance;
- chose the research question;
- wrote the disguised twin for the PD and checked all prompts;
- wrote every pre-registered prediction;

- made the design decisions recorded in the pre-registrations;
- hand-coded the reasoning samples used for validation;
- proved Theorems 1–3 as guided exercises.

Claude:

- wrote the software;
- ran the data collection;
- implemented the pre-registered analyses;
- proposed the mechanism results in Section 6 and verified them numerically;
- proposed the Stage 5 design, which the author adopted;
- drafted the text of this paper.

All claims were checked against the code and data, which are available.

Data and code availability

Code, prompts, seeds, raw answers and analysis outputs are available at [TBD: public repository URL]. The pre-registrations are at osf.io/2uqpj, osf.io/6hz8d and osf.io/cqnsy.

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A Deviations from the pre-registrations

1. **PD scoring error in practice runs:** named answers were scored against the disguised labels. It was fixed before Stage 2.
2. **Gemini 3.8 Flash replaced** by Gemini 3.5 Flash Lite, because of a 20-request daily cap.
3. **Empty replies** (bursts of "!!!!") are re-asked, not scored as unreadable.
4. **gpt-oss's daily token cap** slows its collection; this does not affect the design.
5. **A 10-minute NVIDIA access outage:** its 4,454 refused requests were relabelled as infrastructure failures and re-asked.
6. **The fraction slip:** recorded as an exploratory observation, and corrected only in the pre-registration-3 comparison.
7. **Block D (reasoning effort) dropped** before any of its data existed, so that all models answer the same questions.

8. **The disguised answer boxes** carried the named game's field names (a limitation).

9. **Ten questions re-asked** that should have been scored as the model's own (1 Kimi Stage 2A; 2 Nemotron Stage 2A; 7 Nemotron Stage 2B traps). The re-asked answers stand. The cases are [TBD: listed by question ID].

10. **The pilot answer form** sent explanatory text that leaked words; it was fixed before Stage 2C.